Introduction to 41525 ”Finite Element Methods”

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Preparatory reading:
Course Notes - Day 1
Chapter 1 in Cook et al.

FEM = Finite Element Methods

- Started in the 1950’s – mechanical engineering
- Introduced in Denmark in the 1960’s
- First taught at DTU around 1980
- Today: everyday tool in the industry

FEA - Popular softwares

- Abaqus
- Ansys
- COMSOL
- Nastran
- Loads of others …

FEA vs. FEM (Analysis vs. Methods)

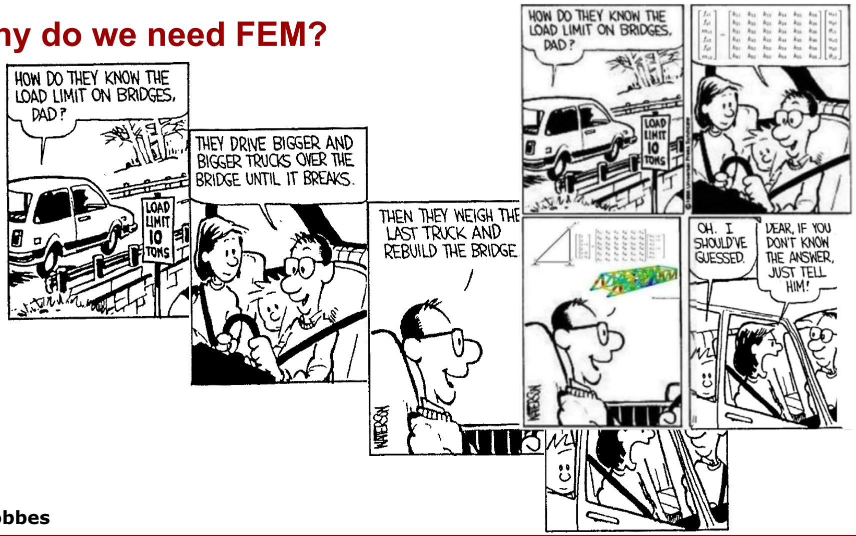
FEA (41812)

- Applied FEA
- Commercial FE codes
- Complex geometries and models
- Linear analysis

FEM (41525)

- Programming the FEM
- Own code development
- Simpler (planar) geometries
- Non-linear analysis

Why do we need FEM?



Calvin and Hobbes

FEM – a smart way to solve PDE's

PDE's Partial Differential Equations

BC's Boundary Conditions

Volume equi.: $\sigma_{ji,j} + \Phi_i = 0 \text{ in } \Omega$ } PDE

Constitutive eq.: $\sigma_{ij} = C_{ijkl}\varepsilon_{kl}$

Strain definition: $\varepsilon_{ij} = \frac{1}{2}(u_{j,i} + u_{i,j})$

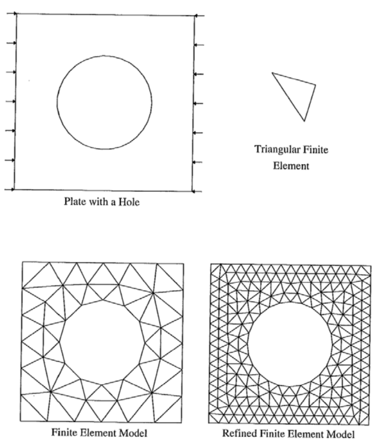
Prescribed displ.: $u_i = u_i^* \text{ on } \Gamma_u$

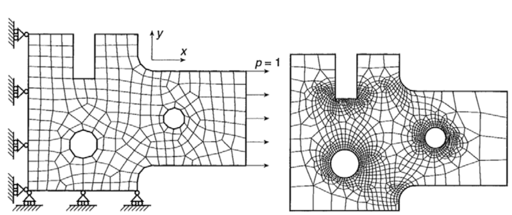
Surface tractions: $\sigma_{ij}n_i = F_j \text{ on } \Gamma_F$ } BC's

Generally no analytical solutions – what to do?

Finite elements

Dividing the domain into finite elements

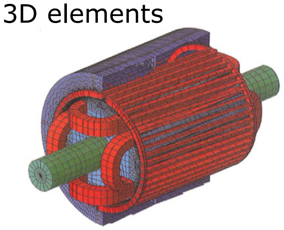


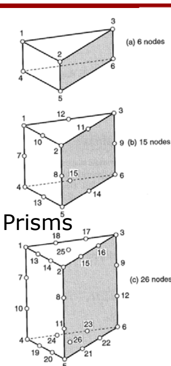


Finer mesh -> better solution !

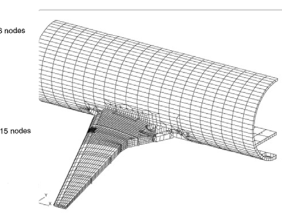
Element types

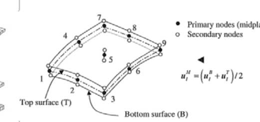
3D elements





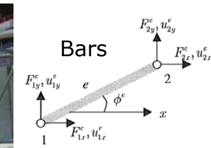
Prisms



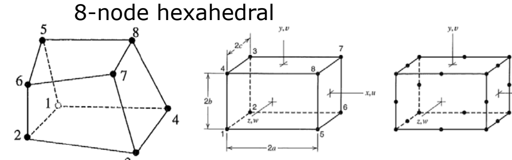


Shells

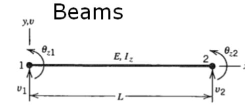
Bars



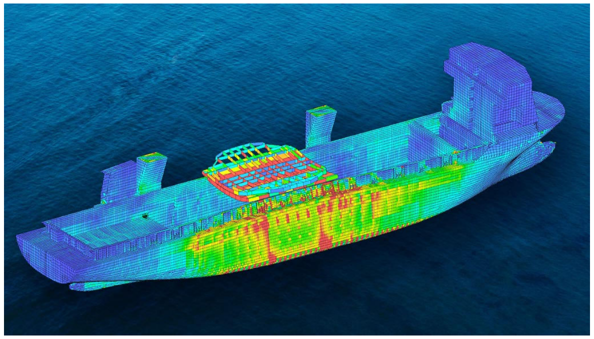
8-node hexahedral



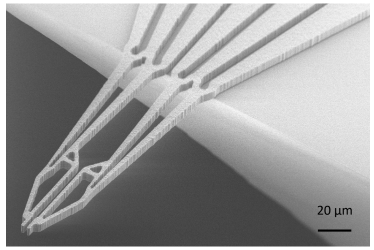
Beams

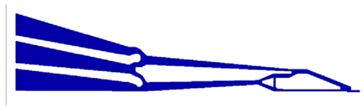


From huge to tiny

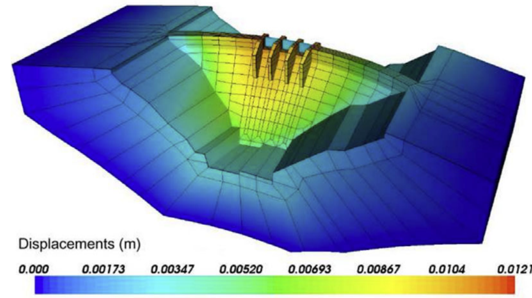


DNV

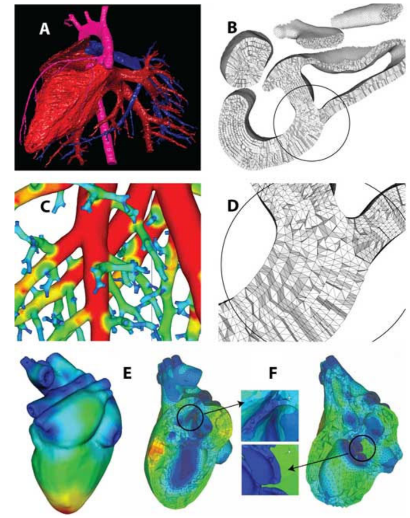




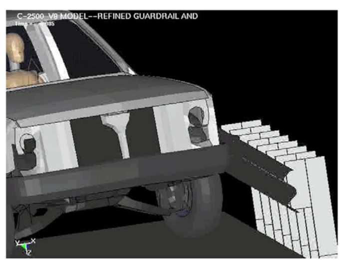
From hard to soft

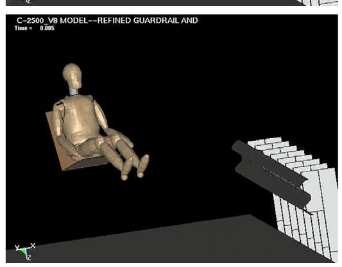


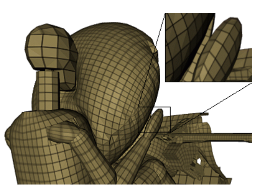
CoBa-group



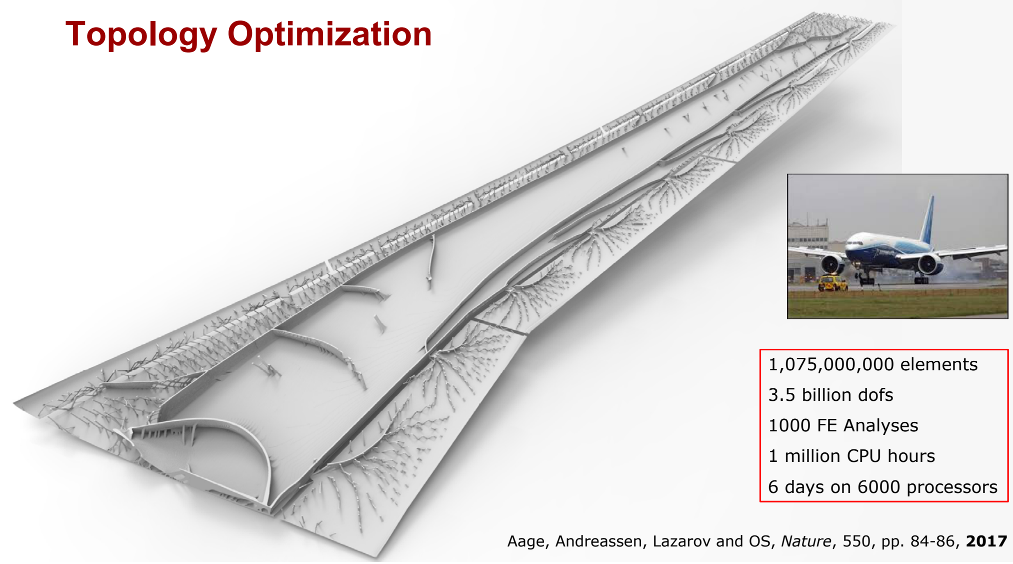
Time dependence







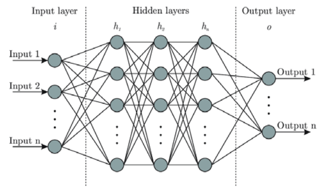
Topology Optimization



1,075,000,000 elements
3.5 billion dofs
1000 FE Analyses
1 million CPU hours
6 days on 6000 processors

Aage, Andreassen, Lazarov and OS, *Nature*, 550, pp. 84-86, **2017**

Do we need FEM when we have AI?



Weak baselines and reporting biases lead to overoptimism in machine learning for fluid-related partial differential equations

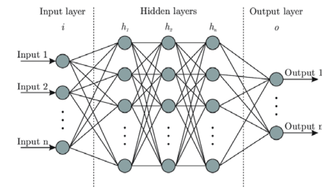
Nick McGreivy[1,2*] and Ammar Hakim[2]

[1*]Department of Astrophysical Sciences, Princeton University, Princeton, New Jersey, USA.
[2]Princeton Plasma Physics Laboratory, 100 Stellarator Rd, Princeton, New Jersey, USA.

arXiv:2407.07218v1

FEM software proven and certified to give correct answers. AI gives no guaranty. The engineer is responsible – not AI!

Do we need <u>to learn</u> FEM when we have AI?



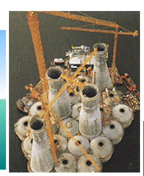
The engineer is responsible!

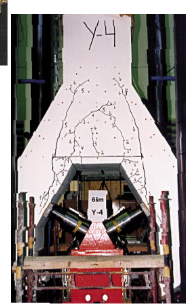
You need to know in depth what is going on

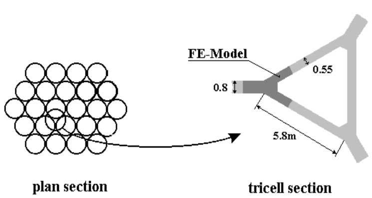
You need to know how to verify results

Be careful! The Sleipner A offshore platform

https://www.ramsay-maunder.co.uk/downloads/nbr06.pdf

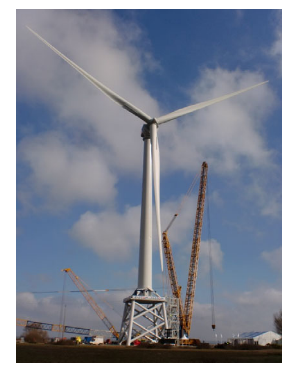






Stress underestimated by 47%, failure postpredicted to 62m, sank at 65m. 57kT, 700mil.$.

Truss structures



Alstom Turbine with LM-73.5

Practicalities at start of course: Finite Element Methods (41525)

Lectures Wednesdays at 8am in room 074, Building 421. Duration: around 2 hours. Smaller follow-up lectures may be given during the day.

Computer exercises 003, 010, 014, 018 in building 421.

Teachers Ole Sigmund (OS), Department of Civil and Mechanical Engineering, Solid Mechanics, Building 404, room 134, e-mail: olsi@dtu.dk

Casper S. Andreasen (CSAN), Department of Civil and Mechanical Engineering, Solid Mechanics, Building 404, room 010, e-mail: csan@dtu.dk

Teaching Assistants Victor Pissinger, email: vmapi@dtu.dk
Jonathan T. Pedersen-Bjergaard, email: jtau@dtu.dk
Brandon Parks, email: bmapa@dtu.dk
Junpeng Wang, email: junwa@dtu.dk

Office hours Office hours will be announced before the assignment periods. Often email requests are more efficient but please do not send same email separately to several TA's.

Apologies Don't miss classes! In case of Force Majeure please send an e-mail to the teachers.

Literature Cook, Malkus and Plesha, "Concepts and Applications of Finite Element Analysis", 4th edition, 2002, John Wiley and Sons. Buy it in the bookstore (Building 101) - see https://polyteknisk.dk/41525-finite-element-metoder-dtu. Handouts during the course.

Technical problems If you have technical problems in the M-Data-bar please contact CONSTRUCT-support (icon on the desktop).

Homework Course notes have one chapter pr. course week. Each chapter also refers to recommended reading in the Cook book. Further info and material may be found on DTU Learn.

Assignments and examination The course evaluation consists of the evaluation of three individualized group reports and a grade can only be awarded if the individual student's contribution to the project can be ascertained. It must be clearly specified for which sections each student has the (main) responsibility. A group project is not deemed to be individualized if the students merely state that they have contributed equally to all sections of the report or the like. Each group will consist of maximum two students. The reports (respectively 4, 5 and a maximum of 15 pages) must demonstrate and discuss solutions of the assignments. The final grade is based on the overall performance the individual student has demonstrated in the three reports.


Brandon


Victor


Junpeng


Jonathan

Assignments 1) Full assignment published on DTU Learn 23/9 (week 4), 1pm. To be uploaded to DTU Learn by 29/9 at 10pm (maximum 4 pages + cover page).
2) Full assignment published on DTU Learn 4/11 (week 9), 1pm. To be uploaded to DTU Learn by 10/11 at 10pm (maximum 5 pages + cover page).
3) Full assignment published in class 11/11 (week 10). To be uploaded to DTU Learn by 7/12 at 1pm (maximum 15 pages + cover page).

Handout found on Learn

AI use in 41525 FEM

DTU embraces use of AI. However, its use must be declared in accordance with DTU and course guidelines. See a.o.

https://student.dtu.dk/en/exam/exam-cheating/dtu-code-of-honour

https://student.dtu.dk/en/exam/exam-cheating

AI cannot be a co-author of your assignment!

While AI tools may be used to support tasks such as coding, data processing, or drafting text, **students remain fully responsible for all aspects of the submitted work** and **must be able to explain, justify, defend and explain the codes, methods, analyses, and conclusions** presented in the report.

As part of the feedback we provide on the reports, we will also assess whether the declared use of AI appears consistent with the submitted work. If possible inconsistencies are identified, we will invite groups to a discussion clarifying the matters and, where relevant, address potential violations of DTU's plagiarism and code of honor policies.

Tentative course schedule

Week	Date	Lecturer	Topic
1	2/9	OS	Matlab truss FE-analysis
2	9/9	CSAN	Matlab truss FE-analysis (material non-linearity)
3	16/9	BNL/CSAN	Matlab truss FE-analysis (geometrical non-linearity)
4	23/9	OS	Matlab truss FE-analysis (Topology Optimization) Publishing of first full report assignment on DTU Learn (1pm)
	29/9		First report (upload no later than 10pm)
5	30/9	OS	Continuum type FE-analysis
6	7/10	CSAN	Continuum type FE-analysis (Equation solving/stress computations/distributed forces)
		CSAN	Feedback to first report
	14/10		FALL BREAK
7	21/10	CSAN	Continuum type FE-analysis (energy principles).
8	28/10	OS	Continuum type FE-analysis (Isoparametric formulation)
9	4/11	CSAN	Dynamics (eigenvalue analysis) Introduction of final projects Publishing of second full report assignment on DTU Learn (1pm)
	10/11		Second report (upload no later then 10pm)
10	11/11	CSAN	Alternative methods and High Performance Computing. Final assignment.
11	18/11	OS	Topology Optimization. Final assignment.
		OS	Feedback to second report
12	25/11	OS	Error estimates + Mesh-generation. Final assignment.
13	2/12	CSAN/OS	Course evaluation. Final assignment.
	7/12		Hand in third report on final assignment before 1pm

Project	Description
1	Plate modelling
2	Geometrical non-linearity
3	Transient problems
4	Plasticity
5	Incompressibility/Stokes flow
6	Special elements
7	Your own idea ...

Assignments

Report 1: 4 pages*
Assignment handed out Day 2/4
Upload report to Learn before Day 5

Report 2: 5 pages*
Assignment handed out Day 7/9
Upload report to Learn before Day 10

Report 3: Max. 15 pages
Assignment handed out Day 10
Upload report to Learn first day of exam period (1pm December 7th)

Group work

- Form groups of **2** (1 and 3 only in absolute ”emergencies”)
- Preferably same available time / similar ambitions in group
- Preferably programming and elasticity experience in all groups
- Optional: use Python instead of Matlab
- Groups may be reformed after first or second report

  (get accept from teachers and organize on or before Day 5 or 10)

DTU rules:

The course evaluation consists of the evaluation of three individualized group reports and *a grade can only be awarded if the individual student’s contribution to the project can be ascertained. It must be clearly specified for which sections each student has the (main) responsibility. A group project is not deemed to be individualized if the students merely state that they have contributed equally to all sections of the report or the like.*

Recommendations

Each Day/Chapter has a number of exercises at the end

Use the exercises for the following purposes:

- as guideline for your work and code implementation
- as means for checking that your code works and produces reasonable results
- as a trigger for learning to interpret and evaluate the numerical results
- as a checklist for what kind of questions you can expect to see in the assignments

It is highly recommended to prepare you codes such that:

- they easily can solve for alternating structures, loads and boundary conditions
- it is easy to generate plots and graphs for visualization and interpretation of the results

Set up a template for your assignment reports (Word, Latex, etc.) and prepare front page, etc.
Exercises marked with a * are voluntary exercises.

Teaching/learning goals

A student who fully satisfies the course goals, will be able to:

- Use the principle of virtual work to set up finite element equations
- Implement the finite element method for truss and continuum problems
- Apply the finite element method to the solution of static and dynamic mechanical problems
- Solve material and geometrically non-linear problems by use of explicit and implicit solution methods
- Optimize simple mechanical structures based on finite element calculations
- Implement iso-parametric finite elements based on numerical integration
- Verify correctness of calculations based on test problems and comparisons with analytical solutions
- Evaluate the quality of a finite element model when the exact solution is unknown
- Evaluate the quality of finite element calculations based on convergence analyses
- Present results of finite element analyses in a transparent, easily accessible and efficient form
- Set up continuous and discrete forms of the principle of virtual work for arbitrary partial differential equations
- Plan a larger finite element programming project and in an independent way suggest and implement test examples.